

Statistical Efficiency of Birth-Death Tree Parameter Estimation

Stat 700

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The Phylodeep Package

Phylodeep Paper is by J. Voznica et al [1]

- ▶ Parameter estimation is difficult and MLE method can be challenging for larger trees.
- ▶ Major components of the paper to address this
 - ▶ CBLV representation of trees
 - ▶ Pre-trained neural networks for pre-specified tree types
- ▶ The goal is to speed up inference

Usage

- ▶ Input: Phylogenetic tree in Newick format
- ▶ Output: Estimated parameters (e.g., birth rate (λ), death rate (μ), R_0)
- ▶ Given a tree, Phylodeep selects the pre-trained model with the highest probability
- ▶ Outputs actually come with uncertainty estimates from a parametric bootstrap

Tree Generation

- ▶ Trees were generated in parallel using the 'treesimulator' package
- ▶ Explored Phylopomp for this, but format is slightly different and packages weren't quickly interoperable

```
1 # Generate a BD tree with 200-500 tips ,
2 # transmission rate of 0.5, removal rate
3 # of 0.25 and sampling probability of 0.3
4 generate_bd --min_tips 200 --max_tips 500 \
5     --la 0.5 --psi 0.25 --p 0.3 \
6     --nwk tree.nwk --log params.csv
```

Maximum Likelihood Estimation

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Phylodeep Methodology

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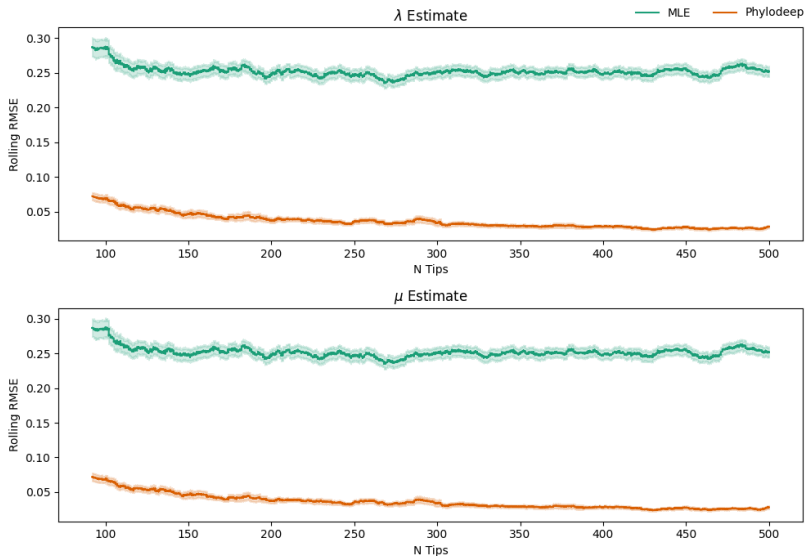


Figure: Rolling bootstrapped RMSE estimates of μ and λ by tree size and resulting 95% CI. Window size is 500.

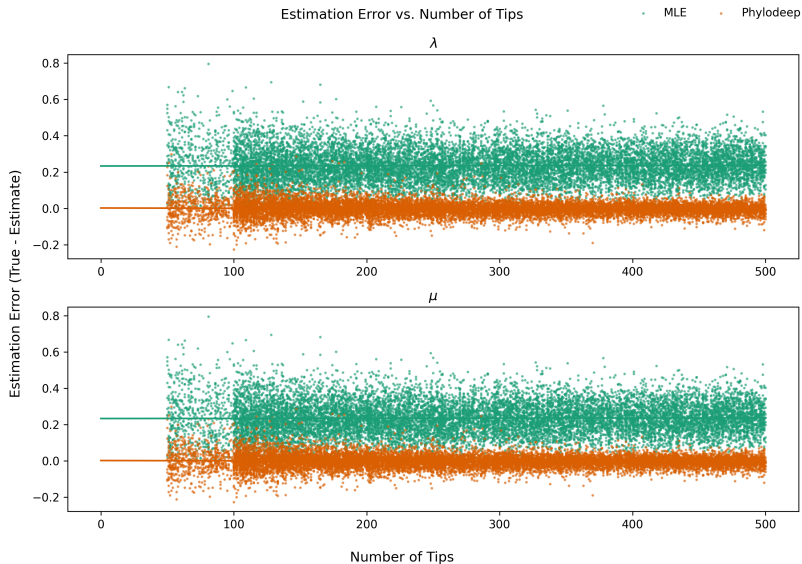


Figure: Raw plot of the absolute errors in λ, μ by tree size.

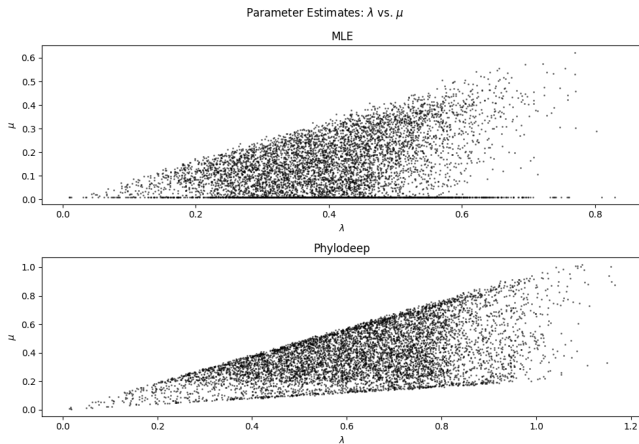


Figure: Plot of λ vs μ estimates to show distribution of estimates.

Main takeaways:

- ▶ RMSE decreases with increasing tree size for both methods.
- ▶ MLE can be used on smaller trees - Phylodeep requires a minimum tip size of around 50.
- ▶ Otherwise, Phylodeep has lower RMSE than MLE, probably due to being trained on 3.9M trees.

References

- [1] J. Voznica et al. “Deep learning from phylogenies to uncover the epidemiological dynamics of outbreaks”. en. In: *Nature Communications* 13.1 (July 2022). Publisher: Nature Publishing Group, p. 3896. ISSN: 2041-1723. DOI: 10.1038/s41467-022-31511-0. URL: <https://www.nature.com/articles/s41467-022-31511-0> (visited on 11/04/2025).